

RISHIT ROY

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SUMMARY

Backend Developer with hands-on experience building production-style REST APIs, distributed systems components, and secure backend architectures using Java and Spring Boot. Skilled in JWT authentication, RBAC, Redis-based rate limiting, Docker, and relational databases via Hibernate/JPA. Passionate about building fault-tolerant, low-latency backend systems with clean architecture and production-grade reliability.

EDUCATION

Noida Institute of Engineering and Technology (NIET)

Aug 2024 – May 2028

B.Tech in Computer Science & Engineering — CGPA: 8.3 / 10 (Till 3rd Semester)

PROJECTS

Distributed API Rate Limiter — [GitHub](#)

2025

Java • Spring Boot • Redis • Docker

- Engineered a distributed rate-limiting library using Spring Boot Filters and Interceptors, enforcing per-user and per-IP request throttling with pluggable algorithm support.
- Implemented **2 rate-limiting algorithms** (Fixed Window Counter and Token Bucket) backed by Redis atomic operations (INCR, EXPIRE, Lua scripts), ensuring thread-safe, sub-millisecond request validation under concurrent load.
- Containerized the Redis instance using Docker, enabling environment-consistent local and staging deployments with zero manual setup.
- Designed the middleware layer to return standardized HTTP 429 responses with retry headers, maintaining API contract integrity without impacting core business logic.
- Achieved clean separation of rate-limiting concerns from application code via Spring's interceptor chain, making the library reusable across **3+ independent API modules**.

Airbnb-Style Hotel Booking Backend System — [GitHub](#) | [Live Swagger](#)

2026

Java • Spring Boot • Hibernate/JPA • PostgreSQL • REST APIs • JWT • Stripe

- Designed a **3-role** backend system (Admin / Host / Guest) with layered architecture (Controller → Service → Repository), supporting hotel listing, room search, and end-to-end booking workflows across **15+ REST endpoints**.
- Implemented transactional booking with optimistic concurrency control to prevent double-booking under simultaneous requests, using JPA-level locking and atomic inventory updates.
- Built dynamic pricing engine with availability-aware room allocation, tracking **3 inventory states** (available, reserved, booked) to ensure accurate conflict-free room assignment.
- Secured all endpoints using JWT-based authentication and RBAC via Spring Security; integrated Stripe payment gateway for booking payment flows.
- Integrated DTO mapping, input validation, global exception handling, and pagination for production-style API hygiene; deployed live on Railway (PostgreSQL on Supabase) — [Swagger UI](#)

TECHNICAL SKILLS

Languages: Java (Primary), SQL

Frameworks & Libraries: Spring Boot, Spring Data JPA, Spring Security, Hibernate, REST APIs

Databases & Caching: PostgreSQL, Redis

Developer Tools: Docker, Git, GitHub, IntelliJ IDEA, Postman, Maven

Backend Concepts: JWT Authentication, RBAC, Rate Limiting, Caching, Concurrency Control, OOP, API Design, Database Modelling, Database Indexing, Exception Handling

ACHIEVEMENTS & CERTIFICATIONS

- Spring Framework Certification (Udemy — Chad Darby, 2025):** Completed 40+ hours covering Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, and REST API development — directly applied in both projects.
- HackerRank Problem Solving Certificate (2024):** Demonstrated proficiency in data structures and algorithm design through standardized assessment.
- LeetCode — 300+ Problems Solved:** Arrays, Graphs, Trees, Dynamic Programming, Backtracking, Greedy, and Hashing — consistently ranking among top performers in timed practice contests.
- Strong algorithmic foundation with emphasis on time/space complexity trade-offs, applied directly in backend concurrency and data-structure design decisions.

INTERESTS

Distributed systems & backend system design • AI/LLM-powered applications & agentic workflows • Exploring new backend tech stacks & open-source backend tooling